



Shore School

2016

Year 11

Preliminary Task 4

Yearly Examination

# Mathematics

## General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board approved calculators may be used
- Answer Questions 1 – 10 on the Multiple Choice answer sheet provided
- Start each of Questions 11 – 14 in a new writing booklet
- In Questions 11–14, show relevant mathematical reasoning and/or calculations
- Write your examination number on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A”
- A BOSTES reference sheet is provided.

**Note:** Any time you have remaining should be spent revising your answers.

Examination Number:

Set:

Total marks – 70

### Section I

Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

### Section II

Pages 6–10

60 marks

- Attempt Questions 11–14
- Allow about 1 hour and 45 minutes for this section

## Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

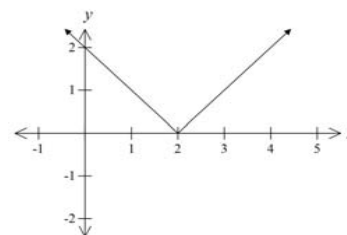
Use the multiple-choice answer sheet for Questions 1–10.

1 What is 0.00412248 written in scientific notation, correct to 4 significant figures?

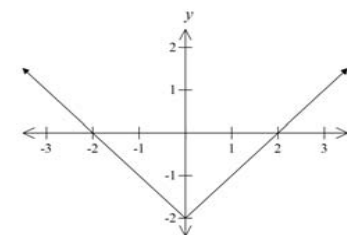
- (A)  $4.1225 \times 10^{-2}$
- (B)  $4.122 \times 10^{-2}$
- (C)  $4.1225 \times 10^{-3}$
- (D)  $4.122 \times 10^{-3}$

2 Which graph best represents  $y = |x - 2|$ ?

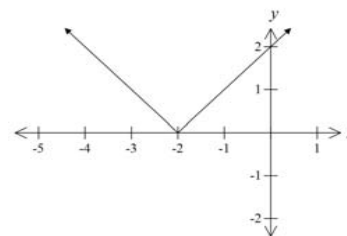
(A)



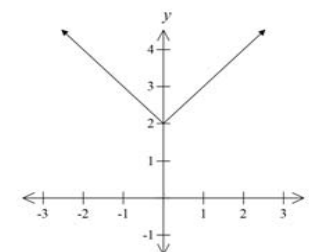
(B)



(C)



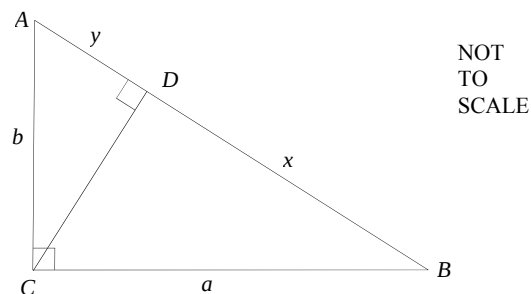
(D)



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- 3 Which of the following is equal to  $\frac{1}{2\sqrt{5}-\sqrt{3}}$  ?
- (A)  $\frac{2\sqrt{5}-\sqrt{3}}{7}$
- (B)  $\frac{2\sqrt{5}+\sqrt{3}}{7}$
- (C)  $\frac{2\sqrt{5}-\sqrt{3}}{17}$
- (D)  $\frac{2\sqrt{5}+\sqrt{3}}{17}$
- 4 Which inequality defines the domain of the function  $f(x) = \frac{1}{\sqrt{x+4}}$  ?
- (A)  $x > -4$
- (B)  $x \geq -4$
- (C)  $x < -4$
- (D)  $x \leq -4$
- 5 What is the gradient of the normal to the curve  $y = 2x^3$  at the point where  $x = 2$  ?
- (A)  $-24$
- (B)  $-\frac{1}{24}$
- (C)  $24$
- (D)  $\frac{1}{24}$
- 6 Which equation represents the line perpendicular to  $2x - 3y - 8 = 0$ , passing through the point  $(2,0)$  ?
- (A)  $3x + 2y - 4 = 0$
- (B)  $3x + 2y - 6 = 0$
- (C)  $3x - 2y + 4 = 0$
- (D)  $3x - 2y - 6 = 0$
- 7 What are the solutions of  $2\cos x = -\sqrt{3}$  for  $0^\circ \leq x \leq 360^\circ$  ?
- (A)  $x = 30^\circ$  and  $330^\circ$
- (B)  $x = 60^\circ$  and  $300^\circ$
- (C)  $x = 150^\circ$  and  $210^\circ$
- (D)  $x = 120^\circ$  and  $240^\circ$
- 8 What is the best description of the nature of the roots of  $3x^2 - 7x + 2 = 0$  ?
- (A) two real, irrational roots
- (B) one real, irrational root
- (C) two real, rational roots
- (D) one real, rational root
- 9 What is the value of  $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2}$  ?
- (A) Undefined
- (B)  $-4$
- (C)  $0$
- (D)  $4$

- 10 Let  $BC = a$ ,  $AC = b$ ,  $BD = x$  and  $AD = y$ . Triangle  $ADC$  is similar to triangle  $ACB$ .



Which of the following statements is correct?

- (A)  $\frac{b}{y} = \frac{a}{x}$
- (B)  $\frac{b}{y} = \frac{x+y}{a}$
- (C)  $\frac{b}{y} = \frac{x+y}{b}$
- (D)  $\frac{b}{y} = \frac{a}{b}$

## Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

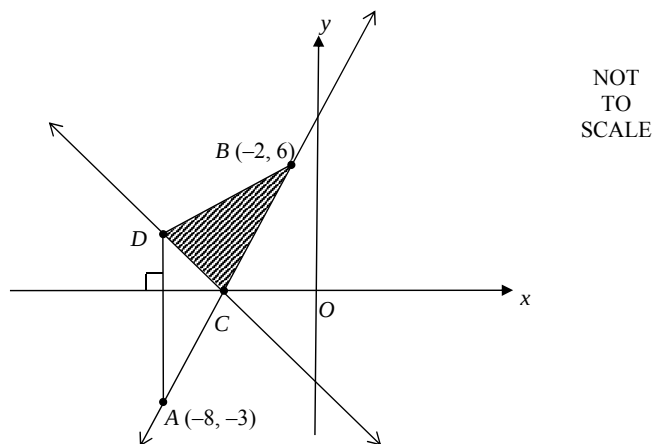
In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

**Question 11** (15 marks) Use a SEPARATE writing booklet

- (a) Solve  $7 - 3x \geq 22$ . 2
- (b) Write, using an algebraic technique,  $0.\dot{3}\dot{2}\dot{4}$  as a fraction in its simplest terms. 2
- (c) Solve  $|x - 3| = 2x - 4$ . 3
- (d) Simplify  $\frac{x^2 - 4x - 5}{4x^2 + 4x + 4} \times \frac{x^3 - 1}{x^2 - 1}$ . 3
- (e) On the number plane shade the region defined by  $x^2 + y^2 < 9$  and  $y \geq x + 3$ . 3
- (f) A regular polygon has an interior angle of  $171^\circ$ .  
How many sides does the polygon have? 2

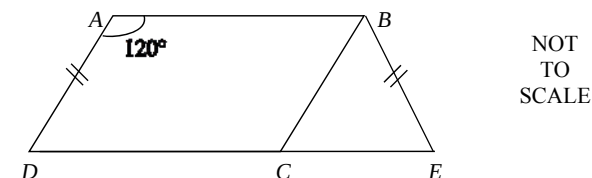
**Question 12** (15 marks) Use a SEPARATE writing booklet

- (a)  $A(-8, -3)$  and  $B(-2, 6)$  are two points on the number plane. The line  $AB$  intersects the  $x$ -axis at  $C$ . The line  $CD$  makes an angle of  $135^\circ$  with the positive  $x$ -axis. The line  $AD$  is parallel to the  $y$ -axis.



- |                                                                                                |   |
|------------------------------------------------------------------------------------------------|---|
| (i) Given the equation of the line $AB$ is $3x - 2y + 18 = 0$ , find the co-ordinates of $C$ . | 1 |
| (ii) Show that the equation of the line $CD$ is $x + y + 6 = 0$ .                              | 2 |
| (iii) Find the co-ordinates of $D$ .                                                           | 1 |
| (iv) Find the exact length of $CD$ , in its simplest form.                                     | 1 |
| (v) Find the perpendicular distance from $B$ to the line $CD$ .                                | 2 |
| (vi) Find the area of $\triangle BCD$ .                                                        | 1 |
|                                                                                                |   |
| (b) Simplify $\cot \theta - \cot \theta \cos^2 \theta$ .                                       | 2 |
| (c) Show that $f(x) = x^3 - x$ is an odd function.                                             | 2 |

- (d)  $ABCD$  is a parallelogram with  $\angle DAB = 120^\circ$ . The side  $DC$  is produced to  $E$  so that  $AD = BE$ .



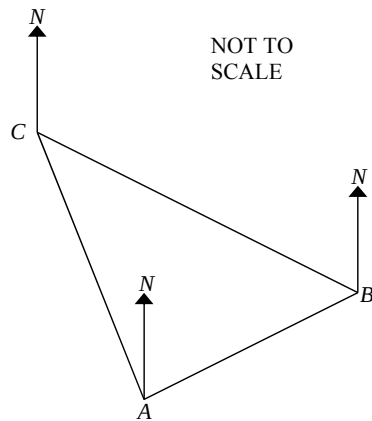
- (i) Copy or trace the diagram into your writing booklet.
- (ii) Prove that  $\triangle BCE$  is an equilateral triangle.

**End of Question 12**

**Question 12 continues on page 8**

**Question 13** (15 marks) Use a SEPARATE writing booklet

- (a) If  $\alpha$  and  $\beta$  are roots of the equation  $2x^2 + 3x - 4 = 0$ , find the value of
- |                                            |   |
|--------------------------------------------|---|
| (i) $\alpha + \beta$                       | 1 |
| (ii) $\alpha\beta$                         | 1 |
| (iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ | 2 |
| (iv) $(\alpha - \beta)^2$                  | 2 |
- (b) Find the values of  $A$ ,  $B$  and  $C$  for which  $2x^2 + 3x + 1 \equiv A(x+2)^2 + B(x+2) + C$ . 3
- (c) A ship sails 50 km from Port A to Port B on a bearing of  $063^\circ$  and then sails 130 km from Port B to Port C on a bearing of  $296^\circ$ .



- |                                                                                                   |   |
|---------------------------------------------------------------------------------------------------|---|
| (i) Copy or trace the diagram into your writing booklet showing all the above information.        | 1 |
| (ii) Show that $\angle ABC = 53^\circ$ .                                                          | 1 |
| (iii) Find the distance from Port A to Port C. Give your answer correct to the nearest kilometre. | 2 |
| (iv) Find the bearing of Port A from Port C. Give your answer correct to the nearest degree.      | 2 |

**Question 14** (15 marks) Use a SEPARATE writing booklet

- (a) Differentiate the following with respect to  $x$
- |                              |   |
|------------------------------|---|
| (i) $y = (3x+4)^5$           | 2 |
| (ii) $y = \frac{x^2+5}{x-2}$ | 2 |
- (b) Differentiate  $y = x^2(3-2x)^4$  with respect to  $x$ . Answer in fully factored form. 3
- (c) Find the equation of the tangent to the curve  $y = x^3 + 2x^2 - 5x$  at the point  $(-3, 6)$ . 2
- (d) A function is defined by  $f(x) = 2x^2 + 7x - 3$ .
- |                                                                         |   |
|-------------------------------------------------------------------------|---|
| (i) Show that $f(x+h) = 2x^2 + 4xh + 2h^2 + 7x + 7h - 3$ .              | 1 |
| (ii) Hence, differentiate $f(x) = 2x^2 + 7x - 3$ from first principles. | 2 |
- (e) The function  $f(x) = \sqrt{4x-1}$  has a tangent with gradient 2 at point  $N$ .
- |                                                            |   |
|------------------------------------------------------------|---|
| (i) Find the co-ordinates $N$ .                            | 2 |
| (ii) Find the equation of the normal to the curve at $N$ . | 1 |

**END OF PAPER**

Section 1

1.  $0.00412248 = 4.122 \times 10^{-3}$  (D)

2. (A)

3. 
$$\frac{1}{2\sqrt{5}-\sqrt{3}} = \frac{1}{2\sqrt{5}-\sqrt{3}} \times \frac{2\sqrt{5}+\sqrt{3}}{2\sqrt{5}+\sqrt{3}}$$
  

$$= \frac{2\sqrt{5}+\sqrt{3}}{4 \times 5 - 3}$$
  

$$= \frac{2\sqrt{5}+\sqrt{3}}{17}$$
 (D)

4.  $f(x) = \frac{1}{1x+4}$

$5x+11 > 0$

$5x > -11$  (A)

5.  $y = 2x^3$

$\frac{dy}{dx} = 6x^2$  at  $x = 2$   
 $= 6(2)^2$   
 $= 24$

Gradient of the tangent = 24

Gradient of the normal =  $-\frac{1}{24}$

as  $m_1 m_2 = -1$  for perpendicular lines. (B)

6.  $2x - 3y - 8 = 0$

$-3y = -2x + 8$

$y = \frac{2}{3}x - \frac{8}{3}$

$m_1 = \frac{2}{3}$

As  $m_1 m_2 = -1$ , for perpendicular lines

$m_2 = -\frac{3}{2}$

$y - y_1 = m(x - x_1)$  (2,0)  $m = \frac{3}{2}$

$y - 0 = \frac{3}{2}(x - 2)$

$2y = -3(x - 2)$

$2y = -3x + 6$

$\therefore 3x + 2y - 6 = 0$  (B)

7.  $2 \cos x = -\sqrt{3}$

$\cos x = -\frac{\sqrt{3}}{2}$

Related angle  $\cos 30^\circ = \frac{\sqrt{3}}{2}$

$\therefore x = 180^\circ - 30^\circ, 180^\circ + 30^\circ$   $\frac{5A}{1C}$

$= 150^\circ, 210^\circ$  (C)

8.  $3x^2 - 7x + 2 = 0$

$\Delta = b^2 - 4ac$

$= (-7)^2 - 4 \times 3 \times 2$

$= 25$

$a = 3, b = -7$

$c = 2$

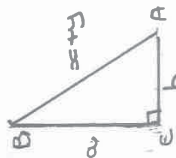
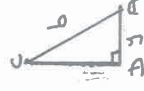
$\therefore$  There are 2 real, rational roots (C)

9.  $\lim_{x \rightarrow -2} \frac{x^2 - 4}{x + 2} = \lim_{x \rightarrow -2} \frac{(x-2)(x+2)}{(x+2)}$

$= \lim_{x \rightarrow -2} (x-2)$

$= -4$

(B)



(C)

$\frac{b}{y} = \frac{x+y}{b}$

Question 11

a)  $7 - 3x \geq 22$

$-3x \geq 15$

$x \leq -5$

b) Let  $x = 0.324$

$= 0.32424 \dots$

$1000x = 324.2424 \dots$

$10x = 3.2424 \dots$

$990x = 321$

$x = \frac{321}{990}$

$x = \frac{107}{330}$

$\therefore 0.324 = \frac{107}{330}$

c)  $|x - 3| = 2x - 4$

$x - 3 = 2x - 4$

$-x = -1$

$x = 1$

$3x = 7$

$x = \frac{7}{3}$

CHECK:

L.H.S. =  $|x - 3|$

$= |1 - 3|$

$= 2$

R.H.S. =  $2x - 4$

$= 2(1) - 4$

$= -2$

L.H.S.  $\neq$  R.H.S.

$\therefore x = 1$  is not a solution

$\therefore x = \frac{7}{3}$

CHECK:

L.H.S. =  $|x - 3|$

$= |2\frac{7}{3} - 3|$

$= \frac{2}{3}$

R.H.S. =  $2x - 4$

$= 2(2\frac{7}{3}) - 4$

$= \frac{2}{3}$

L.H.S. = R.H.S.

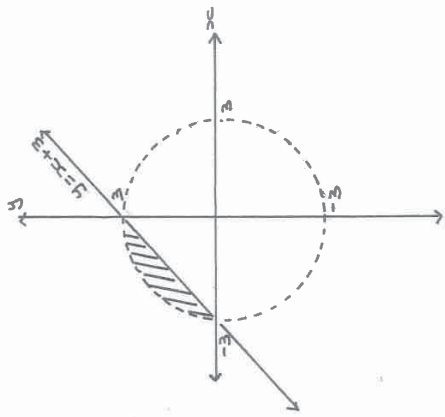
$\therefore x = 2\frac{7}{3}$

is a solution

d)  $\frac{x^2 - 4x - 5}{4x^2 + 4x + 4} \times \frac{x^3}{x^2 - 1}$

$= \frac{(x-5)(x+1)}{4(x^2+x+1)} \times \frac{(x-1)(x^2+x+1)}{(x-1)(x+1)}$

$= \frac{x-5}{4}$



Test (0,0) in  $x^2 + y^2 < 9$  is true

$\therefore$  Shade inside the circle

Test (0,0)  $y > x + 3$

$0 > 0 + 3$  is false

$\therefore$  Shade on the other side.

f) exterior =  $180^\circ$  - interior angle

$= 180^\circ - 171^\circ$

$= 9^\circ$

$= 9^\circ$

number of sides =  $\frac{360}{9}$

$= 40$

OR. interior angle =  $\frac{(n-2) \times 180^\circ}{n}$

$171 = \frac{(n-2) \times 180}{n}$

$171n = 180n - 360$

$-9n = -360$

$n = 40$

### Question 12

a)  $3x - 2y + 18 = 0$  at C,  $y = 0$   
 $3x = -18$   
 $x = -6$

∴ The co-ordinates of C are

$(-6, 0)$

b)  $m_{CD} = \tan 135^\circ$

$= -1$

$y - y_1 = m(x - x_1)$  C  $(-6, 0)$

$y - 0 = -1(x + 6)$

$y = -x - 6$

∴  $x + y + 6 = 0$

iii) at D,  $x = -8$

$x + y + 6 = 0$

$-8 + y + 6 = 0$

$y = 2$

∴ D is  $(-8, 2)$

iv)  $CD = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$   
 $= \sqrt{(-6 + 8)^2 + (0 - 2)^2}$   
 $= \sqrt{8}$

∴  $CD = 2\sqrt{2}$

v)  $d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$  B  $(-2, 6)$   
 $= \left| \frac{-2 + 6 + 6}{\sqrt{1^2 + 1^2}} \right|$

$= \frac{10}{\sqrt{2}}$

$= \frac{10\sqrt{2}}{2}$

∴  $d = 5\sqrt{2}$

v)  $A = \frac{1}{2}bh$   
 $= \frac{1}{2} \times 2\sqrt{2} \times 5\sqrt{2}$   
 $= 10$

∴ Area is  $10 \text{ units}^2$

b)  $\cot \theta - \cot \theta \cos^2 \theta = \cot \theta (1 - \cos^2 \theta)$

$= \cot \theta \times \sin^2 \theta$

$= \frac{\cos \theta}{\sin \theta} \times \sin^2 \theta$

$= \cos \theta \sin \theta$

c)  $f(x) = x^3 - x$

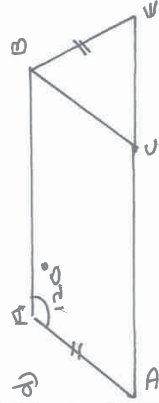
$f(-x) = (-x)^3 - (-x)$

$= -x^3 + x$

$-f(x) = -(x^3 - x)$

$= -x^3 + x$

As  $f(-x) = -f(x)$  the function is odd.



$\angle BCD = 120^\circ$  ∴ opposite angles of a parallelogram are equal

$\angle BCE = 180^\circ - 120^\circ = 60^\circ$  a straight angle is  $180^\circ$

$AD = BC$  ∴ opposite sides of a parallelogram are equal

$\angle BCE = \angle BEC = 60^\circ$  In an isosceles triangle equal angles are opposite equal sides

$\angle BEC = 180^\circ - 60^\circ - 60^\circ = 60^\circ$  angle sum of a triangle is  $180^\circ$

∴  $\triangle BCE$  is an equilateral triangle as all angles are  $60^\circ$

### Question 13

a)  $2x^2 + 3x - 4 = 0$

$a = 2, b = 3, c = -4$

i)  $\alpha + \beta = \frac{-b}{a}$

$= \frac{-3}{2}$

ii)  $\alpha\beta = \frac{c}{a}$

$= \frac{-4}{2}$

$= -2$

iii)  $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta}$

$= \frac{-\frac{3}{2}}{-2}$

$= \frac{3}{4}$

iv)  $(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta$

$= \left(\frac{3}{2}\right)^2 - 4 \times -2$

$= 10\frac{1}{4}$

b)  $2x^2 + 3x + 1 = A(x+2)^2 + B(x+2) + C$

$= Ax^2 + 4Ax + 4A + Bx + 2B + C$

$= Ax^2 + (4A+B)x + (4A+2B+C)$

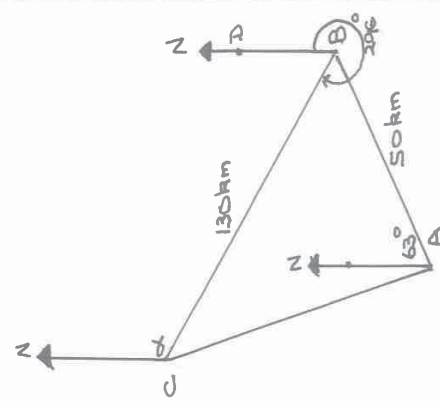
∴  $A = 2$   $4A+B = 3$   $4A+2B+C = 1$

$4(2)+B = 3$   $8+B = 3$   $B = -5$

$-2+(-5)+C = 1$   $-7+C = 1$   $C = 8$

∴  $A = 2, B = -5, C = 8$

c)  $\therefore$  Bearing of A from C is  $116^\circ + 22^\circ = 138^\circ$



$\angle ACB = 360^\circ - 296^\circ = 64^\circ$  ∴ angles about a point add up to  $360^\circ$

$\angle DBA = 180^\circ - 63^\circ = 117^\circ$  ∴ interior angles add up to  $180^\circ$  if the lines are parallel

$\angle ABC = 117^\circ - 64^\circ = 53^\circ$

ii)  $b^2 = a^2 + c^2 - 2ac \cos B$   
 $= 130^2 + 50^2 - 2 \times 130 \times 50 \cos 53^\circ$   
 $= 11576.4047$

$b = 107.5937019$

∴ AC is 108 km to nearest km

iii)  $\cos C = \frac{a^2 + b^2 - c^2}{2ab}$

$= \frac{130^2 + 107.6^2 - 50^2}{2 \times 130 \times 107.6}$

$= 0.92857 \dots$

$C = 21.7856 \dots$

$= 22^\circ$  to the nearest deg.

$\alpha = 180^\circ - 64^\circ - 116^\circ = 116^\circ$  ∴ interior angles are equal if the lines are parallel

∴ Bearing of A from C is  $116^\circ + 22^\circ = 138^\circ$

### Question 14

a) i)  $y = (3x+4)^5$

$$\frac{dy}{dx} = 5(3x+4)^4 \times 3$$

$$= 15(3x+4)^4$$

ii)  $y = \frac{x^2+5}{x-2}$

$$u = x^2+5 \quad v = x-2$$

$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 1$$

$$\frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

$$= \frac{(x-2)2x - (x^2+5) \times 1}{(x-2)^2}$$

$$= \frac{2x^2 - 4x - x^2 - 5}{(x-2)^2}$$

$$= \frac{x^2 - 4x - 5}{(x-2)^2}$$

$$= \frac{(x-5)(x+1)}{(x-2)^2}$$

b)  $y = x^2(3-2x)^4$

$$u = x^2 \quad v = (3-2x)^4$$

$$\frac{du}{dx} = 2x \quad \frac{dv}{dx} = 4(3-2x)^3 \times (-2)$$

$$= -8(3-2x)^3$$

$$\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$$

$$= (3-2x)^4 \times 2x + x^2 \times (-8(3-2x)^3)$$

$$= 2x(3-2x)^4 - 8x^2(3-2x)^3$$

$$= 2x(3-2x)^3[3-2x - 4x]$$

$$= 2x(3-2x)^3(3-6x)$$

$$= 6x(3-2x)^3(1-2x)$$

c)  $y = x^3 + 2x^2 - 5x$

$$\frac{dy}{dx} = 3x^2 + 4x - 5 \quad \text{at } x = -3$$

$$= 3(-3)^2 + 4(-3) - 5$$

$$= 27 - 12 - 5$$

$$= 10$$

$y - y_1 = m(x - x_1)$

$y - 6 = 10(x - (-3))$

$y - 6 = 10x + 30$

$y = 10x + 36$

d)  $f(x) = 2x^2 + 7x - 3$

i)  $f(x+h) = 2(x+h)^2 + 7(x+h) - 3$

$$= 2(x^2 + 2xh + h^2) + 7x + 7h - 3$$

$$= 2x^2 + 4xh + 2h^2 + 7x + 7h - 3$$

ii)  $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{2x^2 + 4xh + 2h^2 + 7x + 7h - 3 - 2x^2 - 7x + 3}{h}$$

$$= \lim_{h \rightarrow 0} \frac{4xh + 2h^2 + 7h}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h(4x + 2h + 7)}{h}$$

$$= \lim_{h \rightarrow 0} 4x + 2h + 7$$

$\therefore f'(x) = 4x + 7$

c)  $f(x) = \sqrt{4x-1}$

$$= (4x-1)^{\frac{1}{2}}$$

$$f'(x) = \frac{1}{2}(4x-1)^{-\frac{1}{2}} \times 4$$

$$= 2(4x-1)^{-\frac{1}{2}}$$

$$= \frac{2}{\sqrt{4x-1}}$$

But  $f'(x) = 2$

$$\frac{2}{\sqrt{4x-1}} = 2$$

$\therefore \sqrt{4x-1} = 1$

$4x-1 = 1$

$4x = 2$

$x = \frac{1}{2}$

$f(x) = \sqrt{4x-1}$

$f(\frac{1}{2}) = \sqrt{4(\frac{1}{2})-1}$

$= 1$

$\therefore$  The co-ordinates of N are  $(\frac{1}{2}, 1)$

ii) gradient of the tangent = 2  
gradient of the normal =  $-\frac{1}{2}$   
as  $m_1 m_2 = -1$  for perpendicular lines.

$y - y_1 = m(x - x_1) \quad (\frac{1}{2}, 1)$

$y - 1 = -\frac{1}{2}(x - \frac{1}{2}) \quad m = -\frac{1}{2}$

$2y - 2 = -x + \frac{1}{2}$

$2x + 4y - 5 = 0$  is the

equation of the normal